Peripheral Pin Connections Guide

This document lists the hardware pin configurations and physical connections for the sensors (nRF coincell transmitter) and the DACs (ESP32 receiver boards).

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1. nRF Coincell Transmitter: SPI Accelerometer Sensors

The nRF5340 board utilizes the `SPIM3` peripheral controller to communicate with both the ADXL345 (low-g) and H3LIS331DL (high-g) accelerometer chips.

The configuration is specified in the device tree overlay file: `coincell-firmware/boards/nrf5340dk_nrf5340_cpuapp`

Pin Connections (nRF5340 -> Sensors)

Signal Name	nRF5340 Pin	ADXL345 (16G)	H3LIS331DL (400G)	Description
SCK (Clock)	`P1.15`	SCL / SCLK	SCL / SCLK	SPI Serial Clock
MOSI (Data In)	`P1.14`	SDA / SDI	SDA / SDI	Master Out Slave In
MISO (Data Out)	`P1.13`	SDO	SDO	Master In Slave Out
CS (H3LIS)	`P1.12`	-	CS	Chip Select (H3LIS331DL)
CS (ADXL)	`P1.11`	CS	-	Chip Select (ADXL345)

> **Note**
>
> - Chip Selects are active low. Only one chip select is pulled low at a time based on the active sensor mode selected.
> - If the range set is $\leq 16g$, the firmware shuts down the H3LIS331DL and activates the ADXL345.
> - If the range set is $> 16g$, the firmware shuts down the ADXL345 and activates the H3LIS331DL.

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2. ESP32-S3 Receiver Boards

There are two alternate receiver codebases depending on which DAC chip you are using.

Option A: SPI DAC (DAC8568) — 16-Bit Precision

Located in: `esp32-receiver`

The ESP32-S3-Zero is connected to a **DAC8568C EVM** evaluation board over a high-speed SPI bus (`SPI2_HOST`).

Pin Connections (ESP32-S3 -> DAC8568)

Signal Name	ESP32-S3 GPIO	DAC8568 Pin	Description
SCLK	`GPIO 7`	SCLK	SPI clock signal (500 kHz)
MOSI	`GPIO 9`	DIN	SPI Master Out (Data Input to DAC)
/SYNC (CS)	`GPIO 13`	/SYNC	Chip Select (Software controlled GPIO)
GND	GND	GND	Common Ground
VCC	3.3V / 5V	VDD	Power Supply

> **Important — EVM Jumper Configuration for DAC8568 EVM (SLAU301)**
>
> - **JP2**: Must be **SHORTED** to connect `LDAC` to GND (allows DAC registers to update outputs instantly).
> - **JP3**: Pins **1-2** shorted (enables the onboard `REF5025` reference regulator).

Option B: I2C DAC (DAC7578) — 12-Bit Precision

Located in: `esp32-receiver-dac7578`

The ESP32-S3-Zero is connected to a **DAC7578** over the I2C bus (`I2C_NUM_0`) running at Fast-mode Plus (1 MHz).

Pin Connections (ESP32-S3 -> DAC7578)

Signal Name	ESP32-S3 GPIO	DAC7578 Pin	Description
SDA	`GPIO 1`	SDA	I2C Data Line (Requires pull-up resistor)
SCL	`GPIO 2`	SCL	I2C Clock Line (Requires pull-up resistor)
GND	GND	GND	Common Ground
VCC	3.3V	VDD	Power Supply

Software Specifications

- **I2C Address**: `0x4C` (Base I2C Address)
- **DAC Channels**:

- **X-Axis**: `0x30` (Channel A)
- **Y-Axis**: `0x31` (Channel B)
- **Z-Axis**: `0x32` (Channel C)